Task 9 & 10 Report - Intelligent Fraud Detection System

1. System integration

FraudPipeline (src/pipeline.py) integrates validation, feature engineering, two models (Keras DNN, logistic regression), SHAP explanations, a nearest-neighbour trust score, human-in-the-loop routing and report generation. A Streamlit app (app.py, pages/) exposes it through six pages and only calls src/. Dataset: 283,726 credit-card transactions, 473 frauds (0.167%); split 70/15/15 stratified (198608/42559/42559); preprocessor fit on train only.

Key finding. Both models detect fraud well in ranking terms: test ROC-AUC 0.951 (DNN) and 0.969 (LR), PR-AUC 0.711 and 0.725 against a fraud base rate of only 0.17%. At its operating threshold the DNN catches 56 of 71 test frauds with 13 false alarms among 42,488 legitimate transactions. The simple logistic regression ranks slightly better than the DNN; the DNN's advantage is higher recall at the chosen threshold. Caveats: only 71 frauds in the test set (wide uncertainty), anonymised PCA features (explanations are not business-readable), two days of 2013 data, a random rather than temporal split, and no customer attributes for a fairness audit.

2. Deliverables inventory

Deliverable	Path
Pipeline	src/pipeline.py
Features / data / training / evaluation	src/features.py, src/data.py, src/train.py, src/evaluate.py
App	app.py, pages/, src/ui.py, .streamlit/config.toml
Models	models/fraud_lr_model.pkl, models/fraud_dnn_model.keras, models/preprocessor.pkl, models/metadata.json
Metrics and CSVs	reports/metrics.json, cost_threshold.csv, latency.json, shap_stability.json, fairness_amount.csv, fairness_time.csv, robustness.csv, drift_baseline.csv, deployment_readiness.csv, testing_report.csv
Figures	reports/figures/
Tests	tests/
Documentation	docs/SRS.md, SDD.md, USER_MANUAL.md, RESPONSIBLE_AI_REPORT.md, DEPLOYMENT_GUIDE.md, TESTING_REPORT.md, FINAL_REPORT.md
Notebook	notebooks/01_eda_and_training.ipynb
Deployment	Dockerfile, scripts/run_all.sh

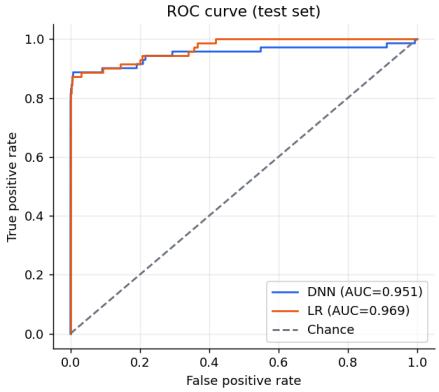
3. Functional testing

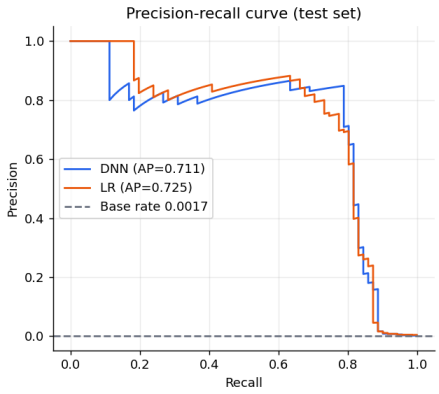
13/13 automated tests pass (valid score, negative amount rejected, missing field rejected, probabilities in [0,1], recommendation in allowed set, non-empty report, batch of 20, batch-vs-single parity, latency). Details: docs/TESTING_REPORT.md.

4. Performance evaluation

Metric	DNN (threshold 0.40)	Logistic regression (0.50)
ROC-AUC	0.951	0.969
PR-AUC	0.711	0.725
F1	0.800	0.752
Precision	0.812	0.870
Recall	0.789	0.662
Accuracy	0.999	0.999
Flag rate	0.16%	0.13%
Confusion matrix (TN/FP/FN/TP)	42475/13/15/56	42481/7/24/47

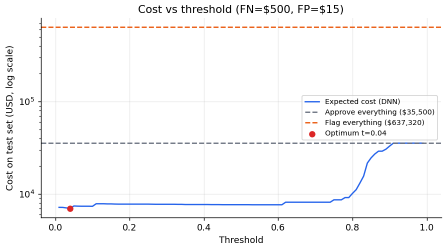
Test set: 42,559 transactions (0.17% fraud, 71 frauds). Source: reports/metrics.json.







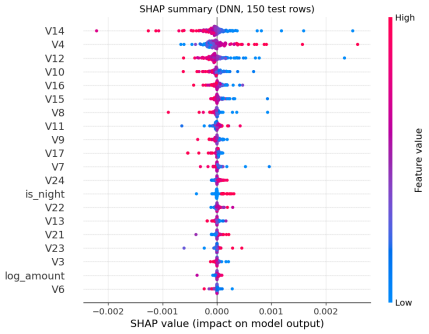
The operating DNN threshold is 0.40: the F1-maximising threshold on the validation split, subject to a validation false-positive-rate guardrail of 30% (the guardrail is not binding here; the unconstrained F1 optimum is also 0.40). With a missed fraud costing $500 and a false alarm $15, the cost-optimal validation threshold is lower, 0.04. On the test set it catches 58 frauds with 34 false alarms (cost $7,010), against 56 frauds and 13 false alarms (cost $7,695) at 0.40. Baselines: approving everything costs $35,500; flagging everything $637,320. The F1 threshold is kept as the operating point because it keeps the alert volume lowest; a bank that prices misses at $500 should consider switching to 0.04. All thresholds are recorded in models/metadata.json.



Latency: single row 7.3 ms mean / 10.0 ms p95 without SHAP; 430 ms mean / 545 ms p95 with SHAP; batch 4,197 rows/s.

5. Explainability

Top-5 SHAP factors per decision; stability 0.995 +/- 0.019 Spearman under 1% noise. Most important features: V14, V4, V12, V10, V16.



6. Responsible AI

No customer attributes exist, so subgroup performance is measured by amount band and time of day: recall gaps 0.311 and 0.265 (criterion 0.20; not met, based on 7-34 frauds per segment). Robustness: no DNN class flips within 3 SD of random perturbation (15/15 cases). Human review: 1,895 of 42,559 test transactions (4.5%) go to MANUAL REVIEW; of the 71 test frauds 14 are auto-blocked, 45 reviewed and 12 missed. Full discussion in docs/RESPONSIBLE_AI_REPORT.md.

7. Deployment readiness

check	criterion	measured	status
DNN predictive power	test ROC-AUC >= 0.70	0.951	PASS
LR predictive power	test ROC-AUC >= 0.70	0.969	PASS
DNN precision-recall (rare class)	test PR-AUC >= 0.50	0.711	PASS
Flag rate sanity	DNN flag rate <= 50%	0.2%	PASS
False-positive rate	DNN FPR <= 30%	0.0%	PASS
Expected cost vs flag-all baseline	DNN cost < flag-everything cost	$7,695 vs $637,320	PASS
Single-row latency (no SHAP)	p95 <= 100 ms	10.0 ms	PASS
Single-row latency (with SHAP)	p95 <= 2000 ms	545 ms	PASS
Batch throughput	>= 100 rows/s	4197 rows/s	PASS
Explanation stability	mean Spearman >= 0.80	0.995 +/- 0.019	PASS
Subgroup recall (amount band)	recall gap <= 0.20	0.311	FAIL
Subgroup recall (time of day)	recall gap <= 0.20	0.265	FAIL
Robustness	median flip distance >= 0.5 std	>= 3.000 std (15/15 never flipped)	PASS
Data drift (train vs test)	max PSI <= 0.25	0.001	PASS
Automated tests	all pass	13/13 passed	PASS
Model artifacts & metadata	all files present, metadata complete	complete	PASS

2 of 16 checks fail (Subgroup recall (amount band), Subgroup recall (time of day)). Predictive power, cost, latency, throughput, stability, robustness, drift, tests and artifacts pass.

8. Limitations

Few positives (71 test frauds); anonymised PCA features (explanations not business-readable); two days of 2013 data and a random split (optimistic metrics, no real drift estimate); no customer attributes for fairness; SHAP explains the DNN only.

9. Future enhancements

Temporal validation (train on day 1, test on day 2) and re-validation on recent traffic; probability calibration; amount-weighted cost (a missed $2,000 fraud costs more than a missed $2 one); gradient-boosting comparison; richer behavioural features (velocity, device, merchant) if non-anonymised data is available; a fairness audit with customer attributes; scheduled PSI monitoring job and an analyst feedback loop; authentication and audit logging.